

## FORLI, Italy

WMO: 161470

Lat: 44.195N

Lon: 12.070E

Elev: 97

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 24.6       | 26.9 | 15.7                          | 12.2 | 32.7 | 19.1 | 14.4 | 33.8 | 17.8                  | 48.5 | 15.0 | 44.4 | 3.3                   | 210  | 0.404 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.4                   | 93.4            | 73.1 | 91.0 | 72.5 | 87.9 | 71.3 | 75.5                | 88.3 | 73.7 | 86.7 | 72.4 | 84.9 | 7.1                  | 80   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 71.6                            | 117.1 | 81.7 | 69.7 | 109.6 | 80.3 | 68.1 | 103.5 | 79.0 | 38.9          | 88.8 | 37.3 | 87.0 | 36.1 | 84.4 | 82.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |     |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-----|-------|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |     |       |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p) |       |  |
| 16.7              | 13.9 | 11.8 | DB                         | 18.9 | 97.7               | 4.1 | 2.6                                                | 16.0 | 99.6       | 13.6 | 101.1      | 11.3 | 102.5      | 8.4 | 104.4 |  |
|                   |      |      | WB                         | 17.5 | 78.2               | 3.8 | 1.9                                                | 14.7 | 79.5       | 12.4 | 80.6       | 10.3 | 81.7       | 7.5 | 83.1  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 57.5   | 39.5  | 41.6  | 49.2  | 54.8  | 63.9  | 71.5  | 75.7  | 75.3  | 67.5  | 59.4  | 49.9  | 40.9 |
|                                                                               | DBStd       | 13.95  | 5.06  | 5.75  | 6.09  | 5.19  | 5.25  | 5.46  | 4.41  | 5.00  | 5.04  | 5.88  | 6.35  | 5.74 |
|                                                                               | HDD50       | 1041   | 328   | 238   | 86    | 17    | 0     | 0     | 0     | 0     | 6     | 79    | 287   |      |
|                                                                               | HDD65       | 3753   | 792   | 655   | 489   | 306   | 84    | 9     | 0     | 31    | 187   | 452   | 748   |      |
|                                                                               | CDD50       | 3788   | 1     | 3     | 62    | 162   | 432   | 644   | 798   | 783   | 524   | 299   | 76    | 4    |
|                                                                               | CDD65       | 1027   | 0     | 0     | 0     | 1     | 51    | 204   | 333   | 319   | 104   | 15    | 0     | 0    |
|                                                                               | CDH74       | 10228  | 0     | 0     | 2     | 23    | 501   | 2030  | 3571  | 3252  | 766   | 83    | 0     | 0    |
|                                                                               | CDH80       | 3890   | 0     | 0     | 0     | 2     | 102   | 719   | 1503  | 1390  | 169   | 5     | 0     | 0    |
| Wind                                                                          | WSAvg       | 3.9    | 3.1   | 3.7   | 4.6   | 4.4   | 4.3   | 4.2   | 4.8   | 4.1   | 3.7   | 3.3   | 3.1   | 3.3  |
| Precipitation                                                                 | PrecAvg     | 26.0   | 1.4   | 1.8   | 2.1   | 2.4   | 2.1   | 1.8   | 1.3   | 1.9   | 2.9   | 3.1   | 3.0   | 2.1  |
|                                                                               | PrecMax     | 36.5   | 3.4   | 6.3   | 4.7   | 4.7   | 5.4   | 4.5   | 4.2   | 4.3   | 6.4   | 6.3   | 6.3   | 6.4  |
|                                                                               | PrecMin     | 16.9   | 0.0   | 0.1   | 0.2   | 0.8   | 0.4   | 0.6   | 0.1   | 0.1   | 0.9   | 0.6   | 0.5   | 0.3  |
|                                                                               | PrecStd     | 5.4    | 1.0   | 1.3   | 1.4   | 1.0   | 1.1   | 1.0   | 0.9   | 1.2   | 1.5   | 1.7   | 1.6   | 1.4  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 59.3  | 61.7  | 71.9  | 77.1  | 87.6  | 93.6  | 96.5  | 97.2  | 88.1  | 80.5  | 70.0  | 60.8 |
|                                                                               |             | MCWB   | 50.4  | 51.3  | 57.7  | 63.1  | 69.1  | 73.3  | 73.7  | 74.0  | 72.3  | 65.7  | 60.6  | 52.2 |
|                                                                               | 2%          | DB     | 55.2  | 58.7  | 67.7  | 72.0  | 82.4  | 90.1  | 93.1  | 94.0  | 84.4  | 76.7  | 65.8  | 55.7 |
|                                                                               |             | MCWB   | 48.2  | 49.5  | 55.5  | 59.4  | 66.5  | 72.3  | 73.1  | 73.3  | 69.7  | 64.4  | 57.5  | 49.6 |
|                                                                               | 5%          | DB     | 50.4  | 55.5  | 64.4  | 69.8  | 79.0  | 87.4  | 90.0  | 91.1  | 81.0  | 73.4  | 62.6  | 52.0 |
|                                                                               |             | MCWB   | 45.1  | 47.5  | 53.9  | 58.3  | 64.3  | 71.5  | 72.0  | 72.4  | 67.4  | 63.1  | 56.5  | 46.8 |
|                                                                               | 10%         | DB     | 48.0  | 52.2  | 61.2  | 66.5  | 76.4  | 84.2  | 87.8  | 87.7  | 78.6  | 69.9  | 59.6  | 49.9 |
|                                                                               |             | MCWB   | 43.8  | 45.9  | 52.2  | 56.0  | 63.2  | 69.1  | 70.9  | 71.2  | 65.6  | 61.1  | 54.8  | 45.8 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 51.7  | 53.7  | 59.5  | 64.0  | 70.5  | 76.6  | 77.2  | 78.7  | 74.3  | 68.3  | 62.8  | 54.3 |
|                                                                               |             | MCDB   | 57.2  | 59.7  | 68.8  | 74.5  | 82.1  | 88.1  | 90.7  | 92.5  | 84.6  | 75.5  | 67.0  | 58.2 |
|                                                                               | 2%          | WB     | 48.8  | 50.6  | 56.7  | 61.0  | 67.9  | 74.2  | 74.9  | 75.7  | 71.6  | 65.9  | 59.4  | 50.7 |
|                                                                               |             | MCDB   | 53.2  | 55.7  | 64.7  | 69.4  | 78.9  | 86.2  | 88.7  | 88.7  | 81.9  | 73.2  | 63.6  | 54.7 |
|                                                                               | 5%          | WB     | 46.2  | 48.8  | 54.9  | 59.4  | 66.0  | 72.2  | 73.4  | 73.6  | 69.4  | 64.1  | 57.5  | 48.4 |
|                                                                               |             | MCDB   | 49.4  | 53.7  | 62.6  | 67.0  | 76.2  | 84.6  | 86.9  | 86.6  | 78.6  | 71.4  | 61.5  | 51.4 |
|                                                                               | 10%         | WB     | 44.0  | 46.8  | 52.9  | 57.6  | 64.4  | 70.3  | 72.1  | 72.1  | 67.2  | 62.3  | 55.3  | 46.1 |
|                                                                               |             | MCDB   | 46.5  | 51.2  | 60.2  | 64.8  | 74.1  | 81.9  | 84.8  | 84.7  | 75.7  | 68.4  | 59.1  | 48.8 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 13.4  | 17.3  | 19.2  | 19.1  | 20.1  | 20.5  | 21.4  | 21.2  | 19.2  | 16.2  | 13.5  | 13.3 |
|                                                                               |             | MCDBR  | 20.1  | 22.7  | 24.4  | 24.4  | 24.6  | 24.5  | 24.5  | 25.5  | 22.6  | 20.6  | 17.0  | 17.5 |
|                                                                               |             | MCWBR  | 14.4  | 15.2  | 14.6  | 14.0  | 12.3  | 11.0  | 11.0  | 11.4  | 11.3  | 11.0  | 11.6  | 12.4 |
|                                                                               | 5% WB       | MCDBR  | 16.9  | 19.7  | 22.0  | 21.1  | 22.3  | 22.6  | 22.3  | 22.6  | 20.9  | 18.7  | 14.1  | 14.2 |
| MCWBR                                                                         |             | 13.0   | 14.2  | 14.4  | 13.2  | 12.2  | 11.1  | 11.2  | 11.3  | 11.5  | 11.1  | 10.9  | 10.9  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.353  | 0.374 | 0.410 | 0.442 | 0.442 | 0.451 | 0.442 | 0.435 | 0.422 | 0.411 | 0.378 | 0.350 |      |
|                                                                               | taud        | 2.397  | 2.309 | 2.221 | 2.155 | 2.202 | 2.224 | 2.256 | 2.293 | 2.302 | 2.308 | 2.336 | 2.425 |      |
|                                                                               | Ebn at Noon | 239    | 254   | 259   | 260   | 264   | 262   | 263   | 260   | 253   | 237   | 225   | 229   |      |
|                                                                               | Edh at Noon | 25     | 32    | 39    | 45    | 45    | 44    | 42    | 39    | 36    | 31    | 26    | 22    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 433    | 703   | 1107  | 1471  | 1833  | 2088  | 2141  | 1835  | 1335  | 837   | 482   | 384   |      |
|                                                                               | RadStd      | 64     | 104   | 146   | 135   | 151   | 112   | 104   | 116   | 90    | 91    | 58    | 49    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (20 neighbors) | +0.72 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -158  | +151  | +81   |

Nomenclature: See separate page